

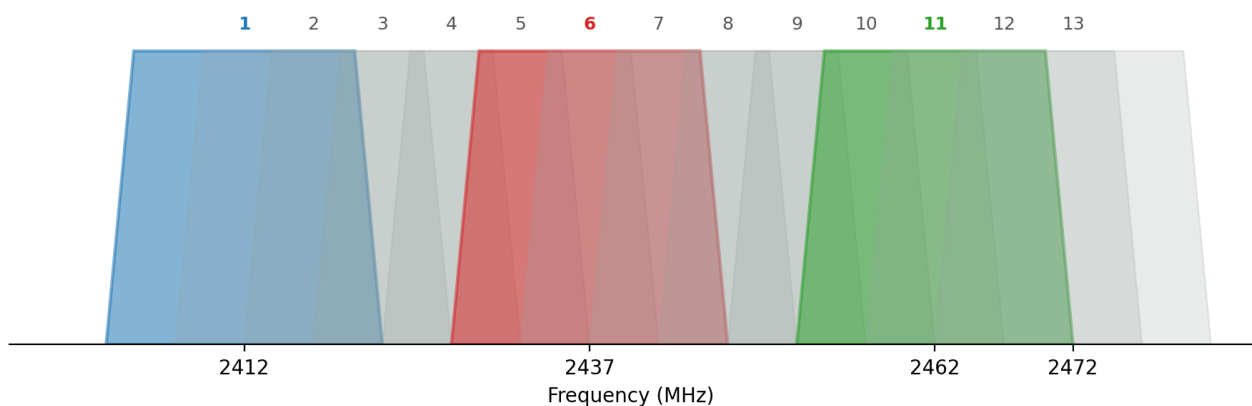
# 2.4 GHz Channel Overlap

Study notes

## Key Facts

- Channels are numbered 1-13 (US: 1-11, Japan also 14). **Centre frequency =  $2407 + 5 \times \text{channel number}$  (MHz)**. Example: channel 6 = 2437 MHz.
- Neighbouring channel numbers are only **5 MHz** apart, but a Wi-Fi channel is **20 MHz wide** (802.11b/DSSS uses about 22 MHz). So channels overlap with their neighbours.
- **Non-overlapping set: 1, 6, 11** (centres 25 MHz apart). Use **20 MHz** channel width on 2.4 GHz.
- The whole 2.4 GHz band is only about 83.5 MHz wide (2400-2483.5 MHz), so there is room for only **three** non-overlapping channels.

2.4 GHz channels (20 MHz wide): 1, 6 and 11 do not overlap



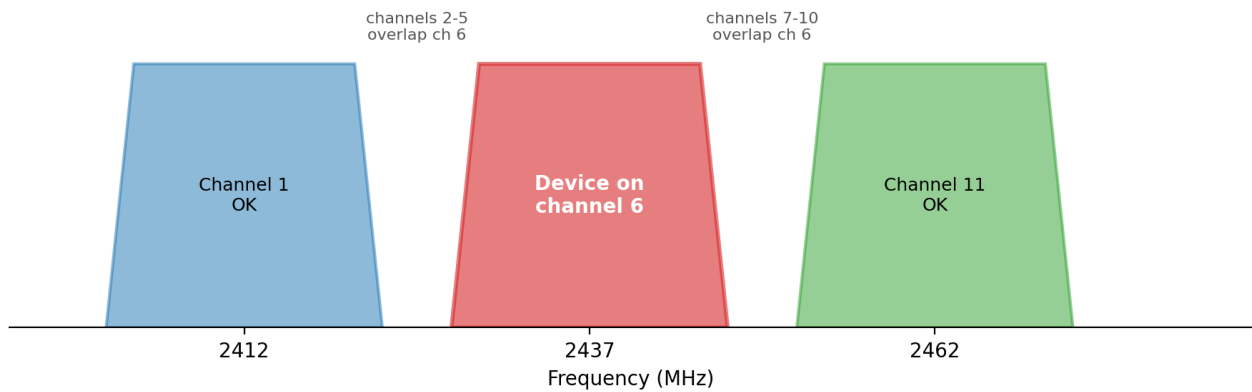
## Example: Which Channels Do Not Overlap with Channel 6?

If one device uses channel 6, a second device must use a different channel and a suitable channel width so that the two do not overlap.

**Use a 20 MHz channel width, and channel 1 or channel 11.**

- Channel 6 at 20 MHz occupies 2427-2447 MHz. Channel 1 occupies 2402-2422 MHz and channel 11 occupies 2452-2472 MHz, so neither touches channel 6.
- Channels 2-5 and 7-10 overlap with channel 6 (partially or at the edge). Do not use them next to channel 6.
- Do not use 40 MHz on 2.4 GHz. It covers about half of the band and overlaps with almost every other channel.

Device on channel 6: use 20 MHz width and put the other device on channel 1 or 11



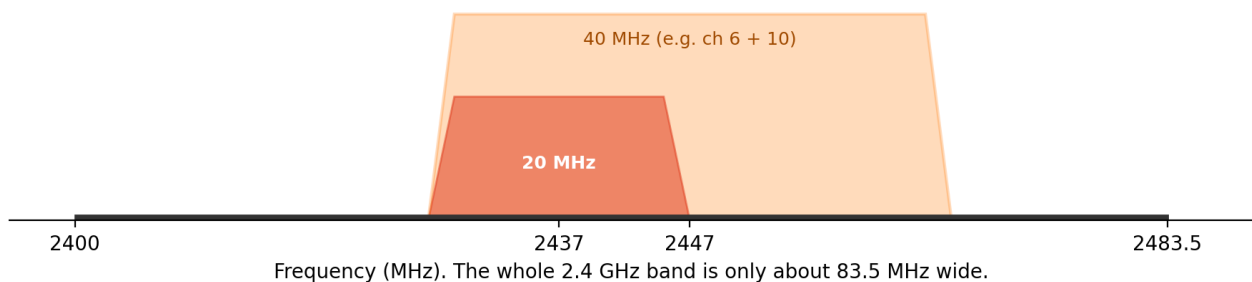
## Channel Table (20 MHz width)

| Channel | Centre (MHz) | Occupies (MHz) | Overlaps with channel 6? |
|---------|--------------|----------------|--------------------------|
| 1       | 2412         | 2402-2422      | No                       |
| 2       | 2417         | 2407-2427      | Edge only (borderline)   |
| 3       | 2422         | 2412-2432      | Yes (strong)             |
| 4       | 2427         | 2417-2437      | Yes (strong)             |
| 5       | 2432         | 2422-2442      | Yes (strong)             |
| 6       | 2437         | 2427-2447      | Same channel             |
| 7       | 2442         | 2432-2452      | Yes (strong)             |
| 8       | 2447         | 2437-2457      | Yes (strong)             |
| 9       | 2452         | 2442-2462      | Yes (strong)             |
| 10      | 2457         | 2447-2467      | Edge only (borderline)   |
| 11      | 2462         | 2452-2472      | No                       |
| 12      | 2467         | 2457-2477      | No                       |
| 13      | 2472         | 2462-2482      | No                       |

Why "edge only" for channels 2 and 10? At exactly 20 MHz they just touch channel 6. In practice the signal skirts and 22 MHz DSSS devices still interfere, which is why the standard recommendation is channels that are 5 numbers apart: **1, 6, 11**. In ETSI regions some designs use 1, 5, 9, 13, but this is a compromise and not the general recommendation.

## Channel Width

A 40 MHz channel uses about half of the whole band



The whole 2.4 GHz band is only about 83.5 MHz wide.

- **20 MHz:** recommended on 2.4 GHz. Three non-overlapping channels (1, 6, 11).
- **40 MHz:** higher peak throughput, but only one usable channel in the band and heavy co-channel interference. Avoid in offices and dense areas. Fine only in very quiet environments.
- On 5 GHz there are many more channels, so wider channels (40/80 MHz) are practical there.

## Quick Rules

- Channels must be **at least 5 numbers apart** (25 MHz) to be safe. Neighbouring numbers (e.g. 5 and 6) overlap almost completely.
- If the given channel is **1**, others are 6 or 11. If it is **6**, others are 1 or 11. If it is **11**, others are 1 or 6.
- Overlapping channels cause **adjacent-channel interference**. Two APs on the **same** channel cause **co-channel interference** (they share airtime), which is often less harmful than partial overlap.
- Other 2.4 GHz noise sources: Bluetooth, microwave ovens, cordless phones, baby monitors.